

Practice Test IV for in Class (on classes 25-28)

Instructions

- I) Write a title, name, date, and course.
- II) Number each problem, circle the number, rewrite the problem, and show all work.
- III) Put a rectangle around your answers.
- IV) Write any fractions in your answers in lowest terms and never leave your answers as improper fractions.

① The ratio of Caleb's weight to Jim's weight is 4 to 5. If their combined weight is 360 pounds, how much does Jim weigh?

Tom has seven barley seeds. The masses of the seven seeds are listed below.

Masses of Mature Barley Seeds (milligrams): 48, 51, 50, 49, 50, 50, 51.

② For the barley seed data above, find the mean mass (round to the nearest thousandth), median mass, mode mass, and mass range.

③ For the barley seed data, find the mass interquartile range and show the distribution of the masses by drawing a dot plot and a box and whisker plot of the data. Then, identify any outliers in the data. If there do not appear to be outliers, write "No outliers."

④ Is the proportion below true?

$$\frac{20}{25} = \frac{15}{20}$$

⑤ If the ratio of red apples to green apples is 2 to 5, and there are 45 green apples, how many red apples are there?

Solve the following proportions.

$$\textcircled{6} \quad \frac{n}{21} = \frac{2}{3}$$

⑦ $\frac{3}{n} = \frac{7}{8}$ ★ Write your answer as a decimal and round to the nearest thousandth.

⑧ Estimate the sum below by rounding each number to the nearest whole number; then add the rounded numbers.

$$203.7 + 53.445$$

⑨ Jesse has 23 football cards and Marty has 18 football cards. Write ratios of these two numbers in six different ways.

⑩ Write the following ratios in lowest terms.

a) 18 to 21 b) 55:66 c) $\frac{73}{81}$ d) 66:55

Weights of Anatolian Shepherd Dogs (pounds): 110, 119, 138, 50

⑪ Using the data above, find the average weight, median weight, mode weight, and weight range.

⑫ Using the same data as problem #11, find the interquartile range and show the distribution of the data with a dot plot and a box and whisker plot. Identify any outliers. If there are no outliers, write "No outliers."

⑬ Estimate the quotient by rounding both numbers to the nearest units place; then multiply the rounded numbers.

$$62.51388 \div 9.499999$$

⑭ Round the following numbers to the nearest ten.

a) 796.034

b) 91.8

c) 99

d) 4

⑮ If a rocket travels 180 miles in two hours, how long would it take for the rocket to travel 270 miles? Write a proportion statement to represent this context and use it to solve the problem.

⑯ If the ratio of the total number of red cherries and yellow cherries to the number of red cherries is 10 to 7, what is the total number of cherries if there are 18 yellow cherries?

⑰ Reduce the following rates to lowest terms.

a) 54 laptops for every 24 desktops.

b) 36 miles in 20 hours.

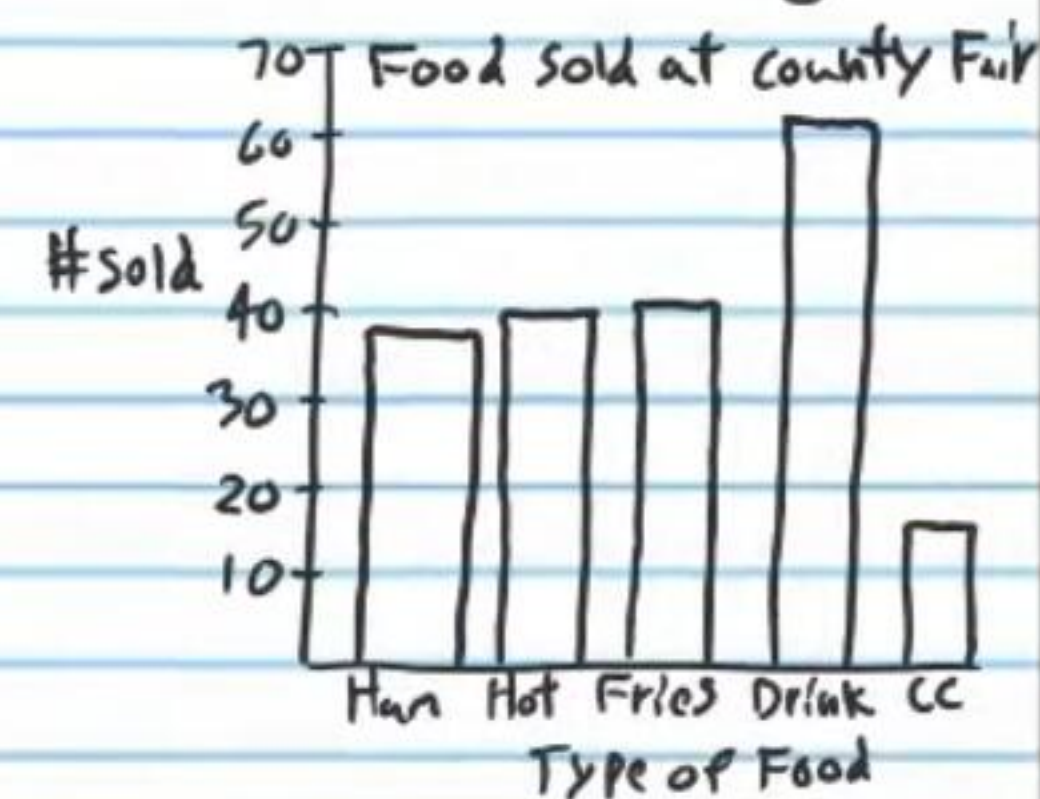
⑱ Find the unit rates for the situations below. You may write your answers as fractions or decimals if necessary.

a) 35 machines built in 7 days. b) 29 yards in 6 minutes.

⑲ Draw a bar graph of the data below. Label the graph properly.

Food Sold at the County Fair

Type of Food	# sold
Hamburgers	38
Hotdogs	40
Fries	41
Drinks	62
Cotton Candy	15



⑳ If it takes 42 days to build eight motorcycles, how many days would it take to build three motorcycles? Find the unit rate and use it to solve this problem.

㉑ If the ratio of pencils to crayons to markers is 8:3:5, and there are 15 crayons, how many pencils are there? How many markers are there?

②② Draw a line graph of the data below. Label the graph properly.

Piano Sales

Year	# Pianos Sold
1993	13
1994	14
1995	10
1996	8
1997	9
1998	5
1999	6
2000	4

②③ Round the numbers below to the nearest tenth.

a) 999.974

b) 80.15

c) 770.50

d) 336

②④ Round the numbers below to the nearest ones place.

a) 0.63

b) 0.05

c) 0.00009843

d) 1.100004

②⑤ Estimate the product below by rounding the first number to the nearest hundred and the second number to the nearest ten; then, multiply the rounded numbers.

$$326.05 \times 58.7$$